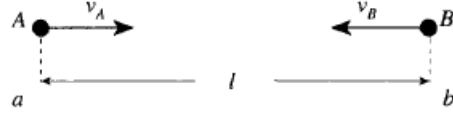


Solution to the Practice Final Examination of **University Physics**

Select five out of following six problems.

1. (20 pts.) A person on Earth observes two rocket ships moving directly toward each other and colliding. At time $t = 0$ in the Earth frame, the Earth observer determines that rocket A, traveling to the right at $v_A = 0.8c$, is at point a , and rocket B is at point b , traveling to the left at $v_B = 0.6c$. They are separated by a distance $l = 4.2 \times 10^8 \text{ m}$.



(a) In the Earth frame, how much time will pass before the rockets collide?

(b) How fast is rocket B approaching in A's frame? How fast is rocket A approaching in B's frame?

(c) How much time will elapse in A's frame from the time rocket A passes point a until collision? How much time will elapse in B's frame from the time rocket B passes point b until collision?

Solution: (a) In the Earth frame

$$t = \frac{b - a}{v_A + v_B} = \frac{l}{v_A + v_B} = 1 \text{ s}.$$

(b) In the Earth frame, $v_x^A = v_A$ and $v_x^B = -v_B$. Then in the A's frame, $u = v_A$,

$$v_x'^B = \frac{v_x^B - u}{1 - v_x^B \frac{u}{c^2}} = \frac{-v_B - v_A}{1 + \frac{v_B v_A}{c^2}} = -0.95c.$$

In the B's frame, $u = -v_B$,

$$v_x'^A = \frac{v_x^A - u}{1 - v_x^A \frac{u}{c^2}} = \frac{v_A + v_B}{1 + \frac{v_A v_B}{c^2}} = 0.95c.$$

(c) In each of the two rocket frames, the time to collision is dilated with respect to that in the Earth frame. So

$$t_A = \frac{t}{\gamma_A} = t \sqrt{1 - \frac{v_A^2}{c^2}} = 0.6 \text{ s},$$

$$t_B = \frac{t}{\gamma_B} = t \sqrt{1 - \frac{v_B^2}{c^2}} = 0.8 \text{ s}.$$

2. (20 pts.) A gas obeys the equation of state

$$p = \frac{T}{V} + \frac{B(T)}{V^2}$$

where $B(T)$ is a function of temperature only. The gas is expanded isothermally at temperature T from the volume V_0 to the volume $V_1 = 2V_0$. The process is reversible.

(a) Find the work done in the expansion.

(b) Find the heat absorbed in the expansion.

Solution: (a) The work done in the expansion is

$$W = \int_{V_0}^{2V_0} p dV = \int_{V_0}^{2V_0} \left(\frac{T}{V} + \frac{B(T)}{V^2} \right) dV = T \ln 2 + \frac{B(T)}{2V_0}.$$

(b) By using the Maxwell relation, we have

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V = \frac{1}{V} + \frac{B'(T)}{V^2}$$

where $B'(T)$ represents the derivative of $B(T)$ with respect to T . Integrating the both side of above equation, we obtain

$$S = \ln V - \frac{B'(T)}{V} + f(T)$$

where $f(T)$ is some function of T . Thus, the heat absorbed in the expansion is

$$\Delta Q = T\Delta S = T(S_1 - S_0) = T \ln 2 + T \frac{B'(T)}{2V_0}.$$

3. (20 pts.) The equation of state for one mole of Van der Waals gas can be written as

$$\left(p + \frac{a}{V^2}\right)(V - b) = RT$$

and the internal energy for the Van der Waals gas is given by

$$U = U_0 + C_V T - \frac{a}{V},$$

where C_V is assumed to be constant.

(a) Determine the adiabatic equation of the one mole of Van der Waals gas.

(b) If the gas expands freely into the vacuum from the volume V_1 to the volume V_2 , what is the temperature change in this process?

Solution: (a) In an adiabatic process

$$dQ = 0.$$

Then,

$$dU = -pdV$$

or

$$C_V dT + \frac{a}{V^2} dV = -\left(\frac{RT}{V - b} - \frac{a}{V^2}\right) dV$$

or

$$\frac{C_V}{R} \frac{dT}{T} = -\frac{dV}{V - b}.$$

Integrating both sides of above equation, after some algebra, we obtain the adiabatic equation

$$T(V - b)^{\gamma-1} = \text{Const.}$$

where $\gamma = (C_V + R)/C_V$. This equation can also be rewritten as

$$\left(p + \frac{a}{V^2}\right)(V - b)^{\gamma} = \text{Const.}$$

In the case of a free adiabatic expansion, the internal energy is unchanged so that

$$\Delta U = C_V(T_2 - T_1) - \left(\frac{a}{V_2} - \frac{a}{V_1}\right) = 0.$$

Therefore

$$(T_2 - T_1) = \frac{a}{C_V} \left(\frac{1}{V_2} - \frac{1}{V_1}\right).$$

4. (20 pts.) If 20 grams of ice at temperature of -10°C is changed into water of 10°C at 1 atm., what is the change of the entropy? (The specific heat of the ice is $c_{pi} = 2.1 \text{ J/g} \cdot \text{K}$ and the specific heat for the water is $c_{pw} = 4.2 \text{ J/g} \cdot \text{K}$. The latent heat absorbed during the ice-water phase transition is $L = 333 \text{ J/g}$.)

Solution: The change of the entropy is

$$\begin{aligned}\Delta S &= \int_{T_0-10}^{T_0} \frac{mc_{pi}dT}{T} + \int_{T_0}^{T_0+10} \frac{mc_{pw}dT}{T} + \frac{mL}{T_0} \\ &= mc_{pi} \ln \frac{273}{263} + mc_{pw} \ln \frac{283}{273} + \frac{mL}{T_0} = 29 \text{ J/K}.\end{aligned}$$

5. (20 pts.) There is a substance of n moles. When the substance expands from a volume V_0 to a volume V at constant temperature T_0 , the work done by it is

$$W = nRT_0 \ln \frac{V}{V_0}.$$

And the entropy of the substance is given by

$$S = nR \frac{V_0}{V} \left(\frac{T}{T_0} \right)^x$$

where x is a fixed constant.

(a) Calculate the Helmholtz free energy of the substance.

(b) Find the equation of state of the substance.

Solution: (a) From $dF = -SdT - pdV$ we have

$$F(T_0, V) - F(T_0, V_0) = -W = -nRT_0 \ln \frac{V}{V_0}$$

and

$$\begin{aligned}F(T, V) - F(T_0, V) &= -\int_{T_0}^T SdT = -\int_{T_0}^T nR \frac{V_0}{V} \left(\frac{T}{T_0} \right)^x dT \\ &= -nRT_0 \frac{V_0}{V} \frac{1}{x+1} \left[\left(\frac{T}{T_0} \right)^{x+1} - 1 \right].\end{aligned}$$

Therefore,

$$F(T, V) = F(T_0, V_0) - nRT_0 \ln \frac{V}{V_0} - nRT_0 \frac{V_0}{V} \frac{1}{x+1} \left[\left(\frac{T}{T_0} \right)^{x+1} - 1 \right].$$

(b) The equation of state is

$$p = -\left(\frac{\partial F}{\partial V} \right)_T = \frac{nRT_0}{V} + \frac{nRT_0 V_0}{V^2} \frac{1}{x+1} \left[\left(\frac{T}{T_0} \right)^{x+1} - 1 \right].$$

6. (20 pts.) A strip of rubber has length l when subject to a tensile force f . If the volume change due to the extension of the strip can be neglected, the differential of the internal energy is then

$$dU = TdS + fdl.$$

(a) Show that

$$\left(\frac{\partial U}{\partial l}\right)_T = f - T\left(\frac{\partial f}{\partial T}\right)_l.$$

(b) Show that the small temperature rise ΔT that takes place in a slow adiabatic stretching of the strip from the natural length l_0 is given by

$$\frac{\Delta T}{T} = \int_{l_0}^l \frac{1}{C_l} \left(\frac{\partial f}{\partial T}\right)_l dl$$

where C_l is the heat capacity of the strip at constant length l .

Solution: (a) By using the Legendra transformation, the Helmholtz free energy is

$$F = U - TS, \quad dF = -SdT + fdl.$$

Thus, we have the Maxwell relation

$$\left(\frac{\partial S}{\partial l}\right)_T = -\left(\frac{\partial f}{\partial T}\right)_l.$$

Viewing entropy S as the function of T and l , we have

$$\begin{aligned} dU &= TdS + fdl \\ &= T\left(\frac{\partial S}{\partial T}\right)_l dT + T\left(\frac{\partial S}{\partial l}\right)_T dl + fdl \\ &= T\left(\frac{\partial S}{\partial T}\right)_l dT + \left[f - T\left(\frac{\partial f}{\partial T}\right)_l\right] dl. \end{aligned} \tag{1}$$

Therefore

$$\left(\frac{\partial U}{\partial l}\right)_T = f - T\left(\frac{\partial f}{\partial T}\right)_l.$$

(b) From the expression of the differential of U , we have the Maxwell relation

$$\begin{aligned} \left(\frac{\partial T}{\partial l}\right)_S &= \left(\frac{\partial f}{\partial S}\right)_l \\ &= \left(\frac{\partial f}{\partial T}\right)_l \left(\frac{\partial T}{\partial S}\right)_l. \end{aligned}$$

From (1), we have

$$C_l = T\left(\frac{\partial S}{\partial T}\right)_l.$$

Then, we have

$$\left(\frac{\partial T}{\partial l}\right)_S = \frac{T}{C_l} \left(\frac{\partial f}{\partial T}\right)_l.$$

Therefore, in the adiabatic (isentropic) process

$$\int_T^{T+\Delta T} \frac{dT}{T} = \ln \frac{T+\Delta T}{T} = \ln \left(1 + \frac{\Delta T}{T}\right) = \int_{l_0}^l \frac{1}{C_l} \left(\frac{\partial f}{\partial T}\right)_l dl.$$

For a small temperature change, we have

$$\ln \left(1 + \frac{\Delta T}{T}\right) \approx \frac{\Delta T}{T}.$$